Set 16: Customer Demographics Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To visualise the age, gender and income profile of customers using a bar chart, a pie chart and a labelled table, and to combine them into an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Customer ID	Age	Gender	Income (in $)
1	28	Female	50,000
2	35	Male	60,000
3	42	Female	75,000

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
customers <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age         = c(28, 35, 42),
  Gender      = c("Female", "Male", "Female"),
  Income      = c(50000, 60000, 75000)
)
write.csv(customers, "Customer_Demographics.csv", row.names = FALSE)  # for Tableau
```

Q1. Bar Chart – Distribution of Customer Ages

> *Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.*

R Program:

```
print(customers[, c("Customer_ID", "Age")], row.names = FALSE)
cat("Mean age:", mean(customers$Age), "| Range:", min(customers$Age), "-", max(customers$Age),
"\n")
p_age <- ggplot(customers, aes(x = factor(Customer_ID), y = Age, fill = Gender)) +
  geom_col(width = 0.55) +
  geom_text(aes(label = paste(Age, "yrs")), vjust = -0.5, size = 4) +
  scale_fill_manual(values = c(Female = "#CC79A7", Male = "#0072B2")) +
```

```
  scale_y_continuous(limits = c(0, 50), breaks = seq(0, 50, 10)) +
  labs(title = "Distribution of Customer Ages", x = "Customer ID", y = "Age (years)", fill =
"Gender") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_age)
ggsave("s16_q1.png", p_age, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Customer_ID Age
           1  28
           2  35
           3  42
Mean age: 35 | Range: 28 - 42
```

Output:

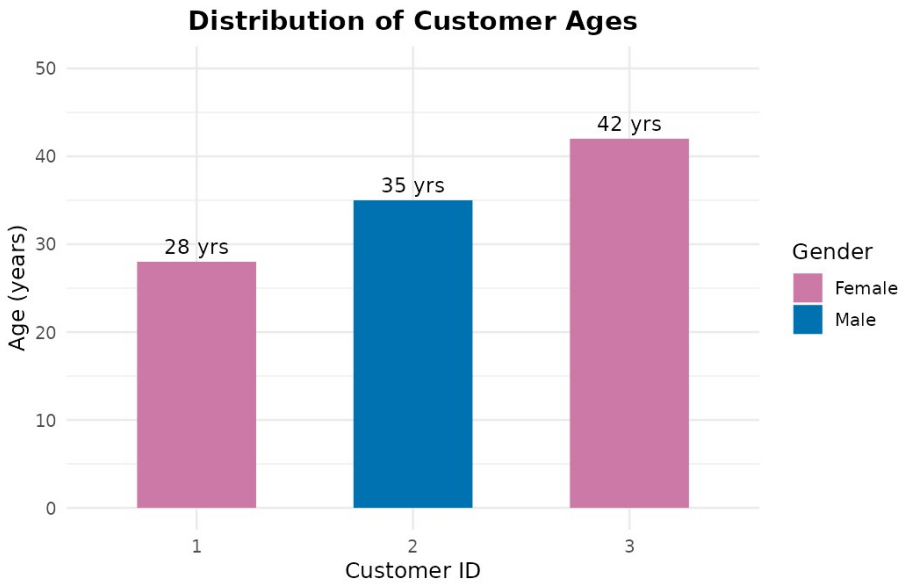


Figure 1: Distribution of customer ages

Inference:

Customer ages range from 28 to 42 years with a mean of 35. Ages rise evenly by 7 years from Customer 1 to Customer 3, so the customers are spread across young and middle-aged adults, with no dominant age group.

Q2. Pie Chart – Customers by Gender

> *Generate a pie chart to represent the distribution of customers by gender.*

R Program:

```
gender_count <- as.data.frame(table(customers$Gender))
names(gender_count) <- c("Gender", "Count")
gender_count$Percent <- round(100 * gender_count$Count / sum(gender_count$Count), 1)
print(gender_count)
p_pie <- ggplot(gender_count, aes(x = "", y = Count, fill = Gender)) +
  geom_col(width = 1, colour = "white") + coord_polar(theta = "y") +
  geom_text(aes(label = paste0(Gender, "\n", Count, " (", Percent, "%)")),
            position = position_stack(vjust = 0.5), colour = "white", size = 4.5) +
  scale_fill_manual(values = c(Female = "#CC79A7", Male = "#0072B2")) +
  labs(title = "Distribution of Customers by Gender", fill = "Gender") +
  theme_void(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_pie)
```

```
ggsave("s16_q2.png", p_pie, width = 5.5, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
  Gender Count Percent
1 Female     2    66.7
2   Male     1    33.3
```

Output:

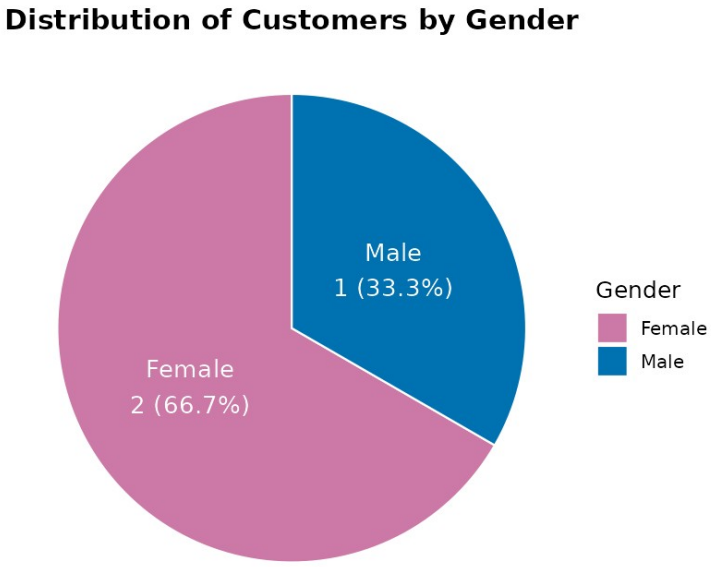


Figure 2: Customers by gender

Inference:

Two of the three customers (66.7%) are female and one (33.3%) is male, so the customer base is female-majority.

Q3. Table – Demographic Information for Each Customer

> *Build a table to show the demographic information for each customer. Label the table elements.*

R Program:

```
tbl <- customers
tbl$Income <- paste0("$", format(tbl$Income, big.mark = ","))
names(tbl) <- c("Customer ID", "Age (years)", "Gender", "Income (in $)")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = tbl_theme)
png("s16_q3.png", width = 900, height = 260, res = 150)
grid.arrange(grid::textGrob("Customer Demographic Information", gp = grid::gpar(fontsize = 14,
fontface = "bold")),
             tg, ncol = 1, heights = c(0.3, 1))
invisible(dev.off())
```

Console Output:

```
 Customer ID Age (years) Gender Income (in $)
           1          28 Female       $50,000
           2          35   Male       $60,000
           3          42 Female       $75,000
```

Table Output:

Customer ID	Age (years)	Gender	Income (in $)
1	28	Female	$50,000
2	35	Male	$60,000
3	42	Female	$75,000

Inference:

Every attribute has a labelled column header with units. Income rises with age: Customer 3 (42 years) earns the most at $75,000 and Customer 1 (28 years) the least at $50,000.

Q4. Tableau Dashboard – Interactive Exploration

> *Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.*

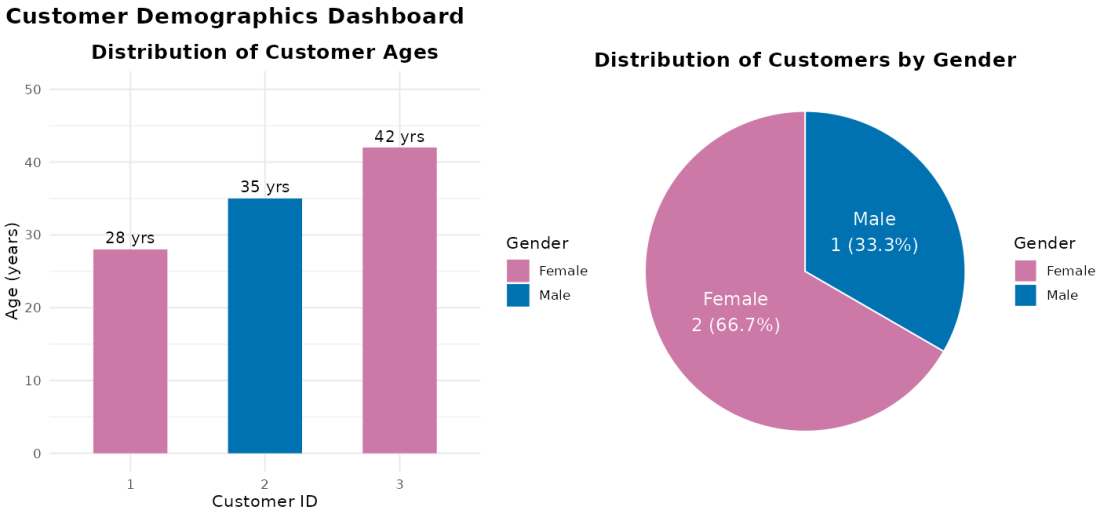
R Program:

```
dash <- (p_age | p_pie) / wrap_elements(full = tg) + plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Customer Demographics Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s16_dash.png", dash, width = 11, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → select **Customer_Demographics.csv** (written by the R program). In the Data Source tab, change **Customer ID** to a **Dimension (String)**.
2. **Sheet 1 – "Age Distribution":** drag Customer ID to **Columns** and Age to **Rows**. Drag Gender to **Color** and Age to **Label**. Title: "Distribution of Customer Ages"; axes: "Customer ID", "Age (years)".
3. **Sheet 2 – "Gender Split":** set the Marks type to **Pie**. Drag Gender to **Color** and **CNT(Customer ID)** to **Angle**. Drag Gender and CNT(Customer ID) to Label and add a **Quick Table Calculation → Percent of Total**.
4. **Sheet 3 – "Customer Table":** drag Customer ID, Gender, Age and Income to **Rows** as discrete fields (or Measure Names / Measure Values to Text). Format Income as Currency ($) and rename the headers.
5. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
6. Drag the sheets onto the canvas: Age Distribution (top-left), Gender Split (top-right), Customer Table (bottom). Add a **Text** object at the top as the dashboard title.
7. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
8. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
9. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Customer_Demographics) so one filter controls every view.
10. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Customer ID	Age (years)	Gender	Income (in $)
1	28	Female	$50,000
2	35	Male	$60,000
3	42	Female	$75,000

Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Clicking the "Female" slice filters the bar chart and table to Customers 1 and 3. Selecting a bar shows that customer's full record, so all three views can be explored together.

Result

The customer demographics were visualised with a labelled bar chart, pie chart and table in R, and combined into an interactive Tableau dashboard. The customers are 66.7% female, aged 28–42, and income increases with age.